Offline semantic bank construction · TRAIN only

TRAIN captions

POS tagging + lemmatization

CLIP text encoder

L2 normalization

Noun bank 5269 × 512

Verb bank 1594 × 512

K = 150, d = 512

Noun aggregator

Verb aggregator

Offline visual feature extraction

Input image I

Predicted boxes producer cache

CLIP vision encoder ViT-B/16

Patch tokens 196 × 768

14 × 14 · CLS removed

ROI pooling N × 768 cache

N × 768

N varies per image (max 87)
Empty boxes → full-frame fallback

Visual projection 768 → 512

N × 512

CSA encoder

T_o

V_0

Object alignment 1
$(V_0, T_o) \rightarrow (V_1, T_1)$

T_r

V_1, T_1

Object alignment 2
$(V_1, T_1) \rightarrow V_o$

V_o

Relation alignment
$(V_o, T_r) \rightarrow V_r$

V_o

V_r

Parallel adaptive decoder (×3)

Linear + softmax

TRAIN: next-token XE

Generated caption

VALID: beam 5 → CIDEr
Best checkpoint selection

BOS / generated prefix
VALID & inference

Embedding + positions

GT captions (shift right)
TRAIN only

CSA adaptation

Cached CLIP ROI + global semantic banks

Frozen Trainable